ANNEALING ALGORITHMS

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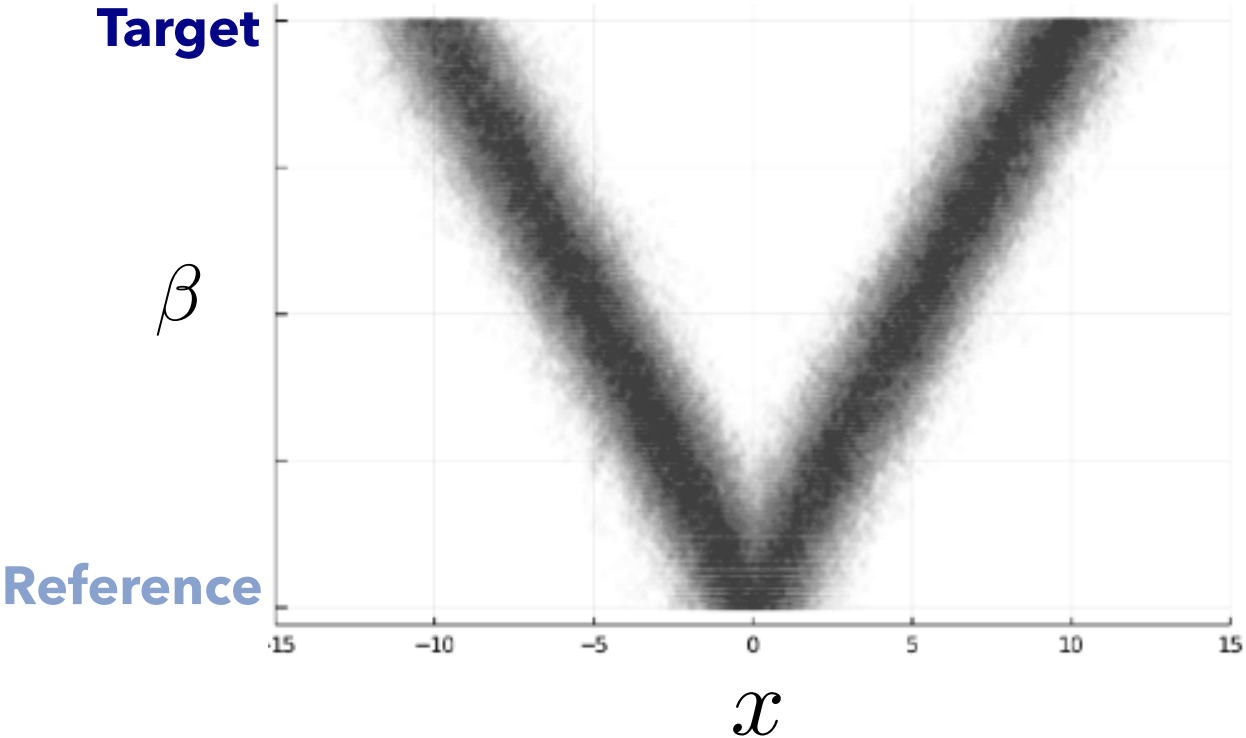
- Annealing is **not** an algorithm but a platform to build algorithms on top of


Target
Reference
β
x
-15
−10
−5
0
5
10
15

- **Input:** An annealing path π_β + a local efficient inference algorithm for each π_β

- **Annealing algorithms** draw inference from the entire path not just the target

- **Output:** a globally efficient inference algorithm for the path $\beta \mapsto \pi_\beta$

- They achieve state-of-the-art performance for hard mult-modal distributions

- Annealing algorithms are meta algorithms:

- Can use the reference to assess the quality and ensure robustness

- Transform a d-dimensional multi-modal target into a $d+1$-dimensional unimodal one

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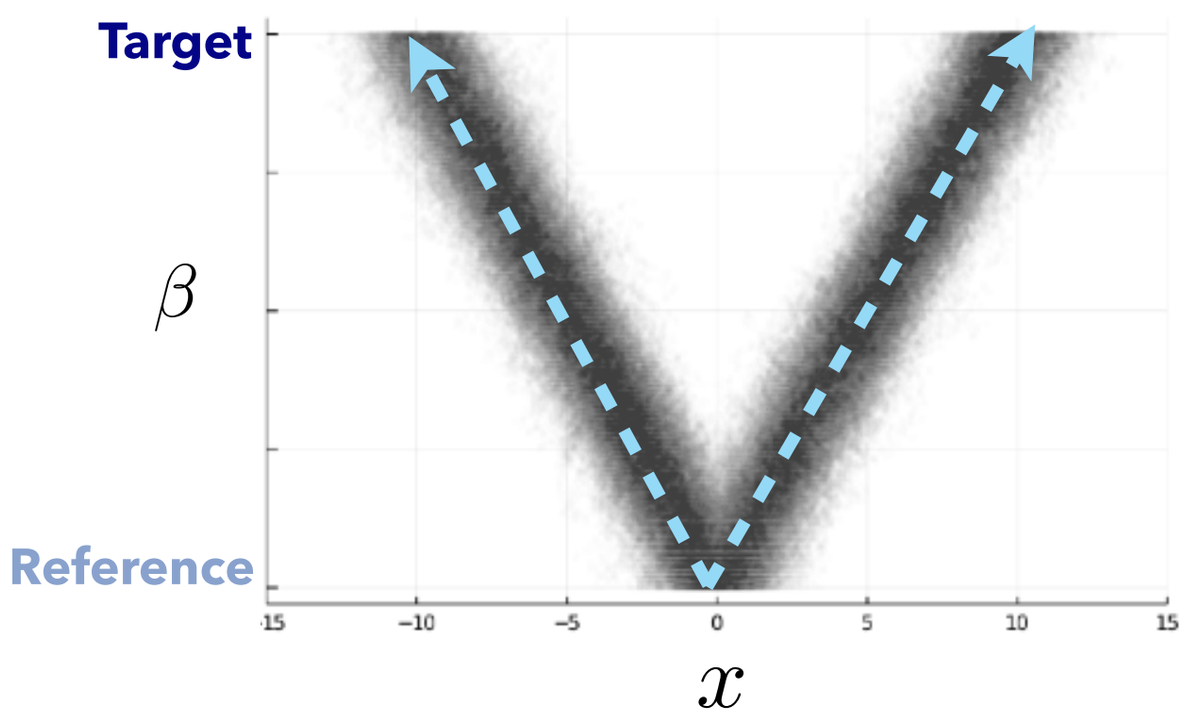

HOW TO CONSTRUCT AN ANNEALING PATH